

IMO 2026 — Autonomous Model Comparison

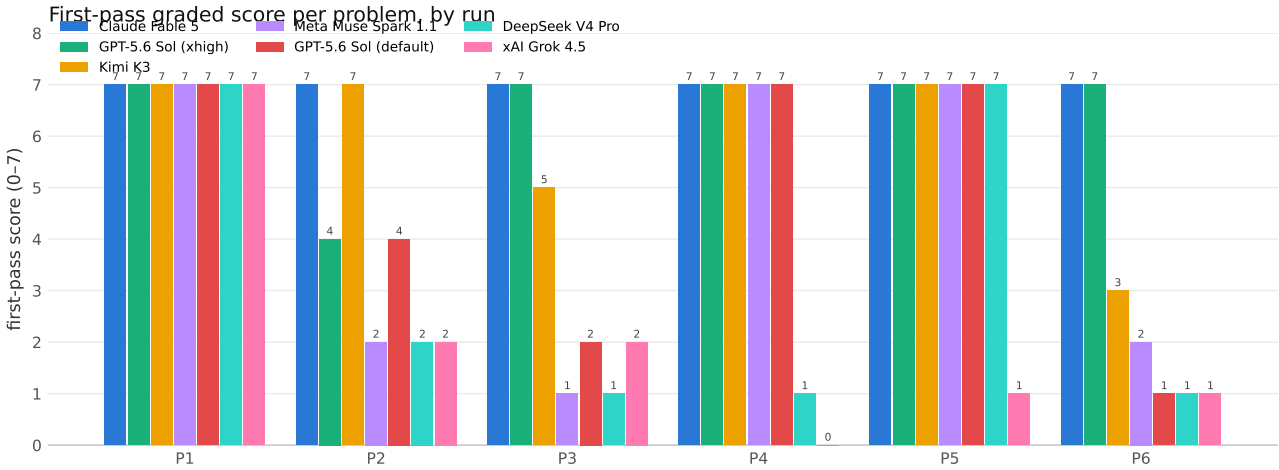
Independently graded results across six runs (five models)

One identical minimal agent harness for every run · IMO 0–7 grading by independent verifier agents, claims checked symbolically and by machine

Final graded scores

Run	P1	P2	P3	P4	P5	P6	Total
Claude Fable 5 (default high)	7	7	7	7	7	7	42/42
GPT-5.6 Sol (xhigh) [†]	7	7	7	7	7	7	42/42
Kimi K3 (default) [†]	7	7	7	7	7	7	42/42
GPT-5.6 Sol (default)	7	4	2	7	7	1	28/42
Meta Muse Spark 1.1 (xhigh)	7	2	1	7	7	2	26/42
DeepSeek V4 Pro (max) [‡]	7	2	1	1	7	1	19/42
xAI Grok 4.5 (xhigh) [‡]	7	2	2	0	1	1	13/42

[†] reached via repair rounds seeded with reviewer defect findings (no external solutions). [‡] Grok and DeepSeek both needed a harness guard to force them to actually call the write-tool (both repeatedly narrated writing files without doing so). Claude Fable 5 is the only run graded 42/42 on its first pass; Muse Spark 1.1 and Grok 4.5 ran single-pass, so their scores are both first-pass and final.



Time and cost (all rounds included)

Run	Final	Total time	Cost (USD)	Output tokens
Claude Fable 5	42/42	2.5 h	\$51.05	705k
GPT-5.6 Sol (xhigh)	42/42	3.8 h	~\$20.54	236k
Kimi K3	42/42	17.4 h	~\$31.40	1,545k
Meta Muse Spark 1.1	26/42	2.3 h	~\$4.45	761k
DeepSeek V4 Pro	19/42	8.9 h	~\$6.05	2,013k
GPT-5.6 Sol (default)	28/42	1.0 h	~\$4.04	80k
xAI Grok 4.5	13/42	3.2 h	~\$14.60	606k

Anthropic API metering for Claude Fable 5; OpenRouter list pricing for GPT-5.6 Sol, Muse Spark 1.1, Grok 4.5; Kimi K3 estimated from logged tokens at list pricing. Times include every retry and repair round.

Findings

- **A frontier tier and a challenger tier.** Claude Fable 5, GPT-5.6 Sol, and Kimi K3 all reach 42/42. The two newest models land far below on first pass — Meta Muse Spark 1.1 at 26/42, DeepSeek V4 Pro at 19/42, xAI Grok 4.5 at 13/42 — all solving the routine problems but none closing the geometry (P2), combinatorics capstone (P3), or number-theory finale (P6).
- **Claude Fable 5 was perfect first-pass**, at its default effort, with no reviewer and no second chances.
- **Effort and feedback lift the frontier models.** GPT-5.6 Sol went 28→39/42 from default to **xhigh**; reviewer-defect feedback then brought GPT (**xhigh**) and Kimi K3 to 42/42. The same levers did not rescue the challengers, whose failures were deeper (wrong answers, unproven cruxes) than the frontier models’ final blemishes.
- **Muse and DeepSeek independently got P3 wrong with the same answer** ($c = (n + 1)/(2n + 1)$, false for all $n \geq 2$, confirmed by exact minimax); DeepSeek also failed to identify P4s answer. **Grok and DeepSeek share a tool-adherence failure**: both ended turns claiming to write files while making zero tool calls, needing a harness guard to force a real write. DeepSeek was also the slowest run by far (8.9 h, 2.0M output tokens — the most compute) yet the cheapest (\$6.05), landing at 19/42.
- **Self-reports inflate.** Almost every run self-reported every problem “solved”; independent grading confirmed far fewer, each deduction backed by a named, checkable defect.

Caveats

- Graders are Claude-based agents, not human medalists.
- All six runs share the identical harness and are directly comparable; reasoning-effort settings are noted per run.
- Full per-turn audit trails (every round of every run) and per-problem grader verdicts with named defects accompany this document; see REPORT.md for attempts/time/tokens/cost.